Harder Geometrical Problems — problem 12.3.2.71

A. Engel — Problem-Solving Strategies · 1998

Let P be an arbitrary point inside $\triangle ABC$; x, y, z the distances of P from A, B, C, respectively ; u, v, w the distances from the sides BC, CA, AB, respectively. The sides of ABC will be denoted by a, b, c, its area by S ; R and r are the radii of circumscribed and inscribed circles. Prove the following inequalities :

(a) $ax + by + cz \geq 4S$; (b) $x + y + z \geq 2(u + v + w)$;

(c) $xu + yv + zw \geq 2(uv + vw + wu)$.